

KeyPad

Arduino and chipKit library support for keypads

Manual



Introduction:

This library is just a quick and easy way to get input through keypads. The library supports keypads with up to 6 columns and up to 5 rows of keys.

You can always find the latest version of the library at <http://www.RinkyDinkElectronics.com/>

For version information, please refer to **version.txt**.

PRE-DEFINED CHARACTERS:

Default characters defined for 3x3, 3x4 and 4x4 keypads:

3x3 Keypad

1	2	3
4	5	6
7	8	9

3x4 Keypad

1	2	3
4	5	6
7	8	9
*	0	#

4x4 Keypad

1	2	3	A
4	5	6	B
7	8	9	C
*	0	#	D

Other-sized keypads have **NO** characters pre-defined. Use `setKeyChars()` to assign characters.

This library is licensed under a **CC BY-NC-SA 3.0** (Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported) License.

For more information see: <http://creativecommons.org/licenses/by-nc-sa/3.0/>

FUNCTIONS:

KeyPad(cols, rows);	
The main class constructor.	
Parameters:	cols: Number of columns of keys (1-6) rows: Number of rows of keys (1-5)
Usage:	KeyPad myKeys(4, 4); // Initialize the library for a 4x4 keypad

setColPins(c0[, c1[, c2[, c3[, c4[, c5]]]]]);	
Configure which pins are connected to the keypad columns.	
Parameters:	c0: Pin connected to keypad column 1 (leftmost) c1: <optional> Pin connected to keypad column 2 c2: <optional> Pin connected to keypad column 3 c3: <optional> Pin connected to keypad column 4 c4: <optional> Pin connected to keypad column 5 c5: <optional> Pin connected to keypad column 6
Usage:	myKeys.setColPins(2, 3, 4, 5); // Setup a keypad with 4 columns connected to pins 2, 3, 4 and 5

setRowPins(r0[, r1[, r2[, r3[, r4]]]]]);	
Configure which pins are connected to the keypad rows.	
Parameters:	r0: Pin connected to keypad row 1 (top) r1: <optional> Pin connected to keypad row 2 r2: <optional> Pin connected to keypad row 3 r3: <optional> Pin connected to keypad row 4 r4: <optional> Pin connected to keypad row 5
Usage:	myKeys.setRowPins(6, 7, 8, 9); // Setup a keypad with 4 rows connected to pins 6, 7, 8 and 9

setKeyChars(row, chars);	
Configure which characters belong to a row of buttons on the keypad.	
Parameters:	row: Which row to set characters for (1 is the upper row) chars: Characters belonging to the row
Usage:	myKeys.setKeyChars(1, "123A"); // Associate the keys on the upper (1 st) row with the characters '1', '2', '3' and 'A' on a keypad with 4 columns

setDebounceDelay(delay);	
Set delay-time to pause after each key press.	
Parameters:	delay: Delay-time in milliseconds
Usage:	myKeys.setDebounceDelay(); // Switch off the delays
Notes:	Set to 0 (default) to switch off.

readKeys();	
Check if a key is being pressed and return the associated character.	
Parameters:	None
Returns:	(char) Character associated with the pressed key. Char(0) is returned if no key is pressed.
Usage:	myKeys.readKeys(); // Check the keys

readRaw();	
Check if a key is being pressed and return the raw button value.	
Parameters:	None
Returns:	(integer) Raw code for the button being pressed. -1 is returned if no button is pressed. The raw code is calculated as (column *10) + row. The third button on the upper row would return 31.
Usage:	myKeys.readRaw(); // Check the keys